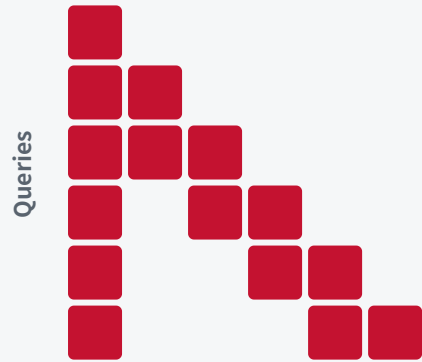


## Fixed pattern

Keys



Fixed  
pattern

Outside pattern /  
future

Input-independent

Visible keys

$$\mathcal{S}_i^{\text{fixed}} = \{j \leq i : i - j < w\} \cup (G \cap [1, i])$$

Predictable cost; the pattern cannot adapt to the query.

## Content-dependent

Keys



Selected  
history

Not selected /  
future

Query-specific

Visible keys

$$\mathcal{S}_i^{\text{dyn}} = \text{TopK}_{j \leq i} g(q_i, k_j)$$

Adaptive retrieval; selector errors exclude relevant keys.